



Fig. 3: Red-transparent conformal coating penetrating open-plated through-holes in an uncontrolled manner

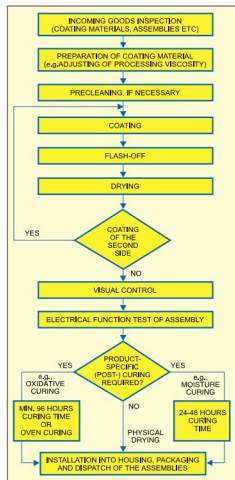


Fig. 4: Flow-chart of the production process

which is also facilitated by the large quantity of coating material originating from the large capacitor located beside the plug.

As a corrective measure, more space could be provided between the plug and capacitor, or the capacitor could be repositioned. If, for example, dipping was chosen for the later coating process, plugs should preferably be located on one side where they simultaneously act as support during the dipping process, and should be arranged in such a way that the assembly can be dipped verti-

TABLE V
Advantages and Disadvantages of Different Coating Application Methods

Method	Advantages	Disadvantages
Brushing	- Little investment - High availability - Suitable for repairs - Suitable for 2-pack coatings - Selective application	- Uneven coating layer - No coating below components - Cannot be automated - Physiologically unsafe, efficient ventilation or separator system may be necessary
Compressed air spraying	- Low investment - High availability - Automatable	- Overspray - Uneven coating layer - No coating below components - High cleaning effort for cabin, tools, etc - Efficient ventilation or separator system may be necessary
Dip coating	- Simultaneous coating of component side and solder side - Coating below components - No overspray - Automatable - Even coating layer - Efficient manufacturing	- High investment costs - Assemblies must be fully submersible
Selective curtain coating/spray coating	- No masking or sealing required - Pointed selective coating - Less consumption of coating material - Even coating layers - Efficient manufacturing	- High investment costs - Single-sided coating - No coating below components
Selective flood/dip coating	- Selective coating possible - Combines advantages of dip coating and the selective curtain/spray coating	Special tools required

cally if possible. A very flat dipping can still be implemented in the coating process, if necessary, but may be subject to special requirements concerning the dip coating unit and/or the coating material. For plugs, in particular, it must be remembered that these can act as so-called thermal bridges, making dewing conditions in the immediate area more likely. This also applies whenever the housing is directly connected with the assembly. A particular design element to be observed when considering selective coating is the open through-plating, or non-plated through holes, in the assembly. The coating materials used have good wetting properties, permitting gaps under components to be easily penetrated. However, this penetration also takes place with open plated-through holes/vias. Two negative effects are possible here. On one hand, this can lead to an uncontrolled lacquer accumulation on the side facing away and, on the other hand, to an 'ab-

duction' of liquid coating material from the coating side. Fig. 3 displays such a reverse side.

Processing of conformal coatings

Conformal coatings can be applied by brushing, spraying, dipping or even by means of automatic selective coating units (refer Table V for advantage and disadvantage corresponding to each method). Not just the conformal coating but all manufacturing steps and also the processing of an assembly should be critically viewed and optimised, in order to ensure its performance for a life cycle as long as possible under the conditions specified. The choice of the base material and the solder resist, the printed circuit board layout and the soldering process have a major influence on the performance of the assembly under climatic stress.

The application of a conformal coating is a decisive process step: